

Tejas Session 9

Tuesday, August 18, 2026 8:41 PM

Factorial :

$$\begin{array}{l} 5! \rightarrow 1 \times 2 \times 3 \times 4 \times 5 \\ 4! \rightarrow 1 \times 2 \times 3 \times 4 \\ 3! \rightarrow 1 \times 2 \times 3 \end{array} \left. \vphantom{\begin{array}{l} 5! \\ 4! \\ 3! \end{array}} \right\} \rightarrow \text{skip}$$

Edge Case : $0! = 1$

Shubham wants to monitor his health by calculating his BMI using Python. He inputs his weight and height to get his BMI, which indicates whether he's underweight, within a healthy range, or overweight. To enhance the program, he adds print statements for a better user experience.

NOTE-

BMI is calculated using weight (in kg) divided by height (in meters) squared. A BMI of 25.0 or more indicates overweight, while 18.5 to 24.9 is considered healthy. This applies to most adults aged 18-65.

input $\rightarrow$ weight
 $\rightarrow$ height

$$BMI = \frac{\text{Weight}}{(\text{height})^2}$$

■ Underweight (<18.5) ■ Normal (18.5-24.9) ■ Overweight (25-29.9) ■ Obese I (30-34.9) ■ Obese II (35-39.9)
■ Obese III (≥40)

$BMI \geq 25 \rightarrow \text{OverWeight}$

$BMI \in [18.5, 24.9] \rightarrow \text{healthy}$

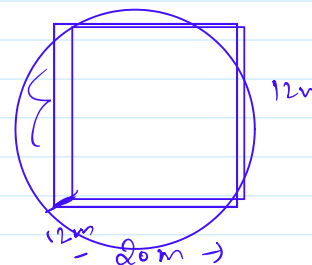
Codeyoung

Inclass Challenge 3- Garden Fencing

Anu recently bought a house and it came with a garden. She absolutely loved the house and wanted to fence the garden. She took a measuring tape and found out that the measurements were in the following order, 20 m, 12 m, and 12 m. Find the cost of fencing at the rate of ₹ 150 per meter.

Create a visually appealing python program that can calculate the price of fencing by calculating the perimeter. Use print statements to make the program look pretty.

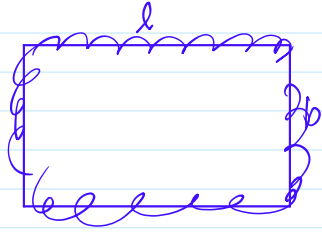
20m
12m
12m
0



150 per meter

↓

Cost



$$\text{per} = 2(l+b) \checkmark$$

↓

$$\underline{150 * \text{per}}$$

Natural No $\rightarrow 1, 2, 3, \dots, 0$

Prime No $\rightarrow$

Composite.

C6 - prime

$$N=5 \leftarrow \text{user}$$

$$1+2+3+4+5=15$$

$$\text{Sum} = N * \frac{(N+1)}{2}$$